# Three Kinds of Sameness: A Complete Classification of Unification Claims

MuningAn · PlanetarySystem · 2026-08 · Paper T1 (first version, with explicit falsification clause)

**Abstract** The term “unified field theory” is used for three logically distinct assertions: the conversion type (one field physically transforms into another), the geometric type (multiple fields are projections of higher-dimensional geometry), and the algebraic type (multiple theories share a deeper algebraic structure). This paper proves that these three exhaust all testable assertions of the form “physical phenomena A and B are the same thing”—sameness claims come in exactly three forms: map, common projection, shared invariant. The proof uses the formal framework of theories (theories as categories, Halvorson 2019) and the formal duality-equivalence literature (De Haro-Butterfield 2018; Weatherall 2018) as its language, and takes the 2025 monograph by De Haro and Butterfield, **The Philosophy and Physics of Duality**, as its principal interlocutor: the monograph formalizes map-type sameness (theoretical equivalence) in its authoritative form, and this paper extends it to the complete trichotomy that also covers the projection and invariant types. Each of the three forms corresponds to a distinct evidence regime (quantitative adjudication / window narrowing / ontological silence), which explains why the three unifications fail in different ways and require different adjudication experiments—this is the formal basis of the adjudication-matrix methodology. The paper includes an explicit falsification clause: if a fourth sameness structure is found, the classification is revised accordingly.**Keywords** unified field theory; theoretical equivalence; category theory; duality; scientific realism

## Introduction

---

The historical narrative of unified field theory has conflated three different assertions into one word, producing a century of confusion. Distinguishing them (see the companion paper **Three Unifications**) is the first step; proving that they **exhaust** the possibilities is the second—the task of this paper. A completeness proof for a classification elevates it from "one viewpoint among many" to "an enumeration of all possibilities"—the qualification required for that classification to serve as a methodological foundation.

## Formal Framework

---

### Theories as categories

We adopt the Halvorson (2019) framework: a physical theory $T$ is a category, whose objects are models of physical systems and whose morphisms are the physically allowed transformations. The "sameness" of two theories then has three basic forms—the standard vocabulary of category theory:

- **Map**: a functor $f : T_A \to T_B$ (possibly an equivalence), translating A's structure into B's structure;
- **Common projection**: there exists a third theory $T_C$ with functors $p_A : T_C \to T_A$, $p_B : T_C \to T_B$ (A and B are both quotients/fibers of C);
- **Shared invariant**: there exists an algebraic/categorical invariant $K$ such that both A and B are representations or realizations of $K$.

The formal duality-equivalence literature (De Haro-Butterfield; Weatherall) chiefly treats the first form (equivalence maps between theories); its framework extends naturally to the latter two.

### Definition of a sameness claim

**Definition**. A "sameness claim" is a proposition $P(A, B)$ supported by observational data: "physical phenomena A and B are the same thing," such that the realization of $P$ must be a computable structure —a concrete relation between the structures of A and B checkable against data.

## Theorem: Completeness of the Trichotomy

---

**Theorem (Completeness of the trichotomy of sameness claims)**. Let $P(A, B)$ be a sameness claim as defined above. Then the structural basis of $P$ must be one of the following three (or a composite):

(1) **Map type**: there exists a map preserving A's observable structure into B (symmetry, duality, or dynamical conversion)—the conversion unification;

(2) **Projection type**: A and B are both projections/restrictions of a common parent structure C—the geometric unification;

(3) **Invariant type**: A and B share an algebraic or kinematic invariant K—the algebraic unification.

A fourth structural basis does not exist.

| Structure | Physical prototype | Evidence regime | Adjudication mode |
|---|---|---|---|
| Map | Electromagnetic duality (self-dual sector); Lorentz transformations | Quantitatively computable | Direct computation (Schwinger inversion) |
| Projection | Kaluza-Klein fibration | Window narrowing | Orbital constraints + nuclear-clock window |
| Invariant | The kinematic algebra of the double copy | Ontological silence | Not adjudicable by computation |

Table 1: Table 1: The one-to-one correspondence of three structures, three evidence regimes, and three adjudication modes.

**Sketch of proof**. The structural basis of a sameness claim is given by the categorical relation between A and B. A basic fact of category theory: any relation between two categories decomposes into composites of three kinds of ingredients—functors (maps), universal constructions (limits/colimits—projections from common domains), and invariants (kernels of representation/module structures). If the realization of $P$ is computable (by definition), its categorical realization must be a constructible functorial relation, hence a composite of the three ingredients. The three ingredients correspond exactly to the three unifications. If a fourth kind, indecomposable into the three, existed, then the realization of $P$ would be non-computable, contradicting the definition. QED (a fully formalized version is left to future work; the present paper's role is to give the structure of the proof and its falsification clause).

## Physical Correspondence: Three Structures ↔ Three Evidence Regimes

---

Each structure determines an evidence regime, explaining the different fates of the three unifications:

This correspondence is the formal basis of the adjudication-matrix methodology: each matrix row is first classified as some sameness claim, and the adjudication channel is then configured according to that class's evidence regime—not a methodological choice, but a structural necessity.

## Relation to De Haro-Butterfield (2025)

---

The 2025 monograph by De Haro and Butterfield, **The Philosophy and Physics of Duality** (Oxford University Press, open access), advances the philosophical formalization of dualities to its authoritative form: theories as structured categories, dualities as equivalence maps, together with a criterion of theoretical equivalence and a duality-based argument for realism. The present theorem stands in a threefold relation to the monograph:

(1) **Containment**. The "sameness" the monograph treats is the first class of our classification (the map type)—they give its most precise modern formalization. We do not duplicate it; we cite it. The formal content of the map type is taken to be the monograph's framework.

(2) **Generalization**. This paper proves that sameness claims come in exactly three classes, of which the map type is only one. The projection type (Kaluza-Klein fibration, AdS/CFT holography) and the invariant type (the kinematic algebra of the double copy) appear in the monograph as examples; here

they are elevated to classification members on a par with the map type—"common projection" and "shared invariant" are sameness structures logically independent of equivalence maps.

(3) **Testability link**. Our structure-evidence correspondence (Table 1) adds a dimension the monograph does not systematize: the structural type of a sameness claim determines how it can be tested—map type by computation, projection type by window narrowing, invariant type by ontological silence. This is the bridge from "formal criteria of theoretical equivalence" to "testing strategies for unification claims," and the point where the adjudication-matrix methodology gains a footing in the philosophy-of-physics literature.

**Methodological statement**: we do not challenge the monograph's equivalence criterion; our claim is that whenever physicists say "A and B are the same thing," the structural basis of the assertion is one of only three kinds, each with a distinct evidential fate—stated within the monograph's framework as a categorical fact (the completeness theorem), with its correction conditions made public through the falsification clause. In venue terms, this paper is addressed to readers of philosophy of physics (SHPSM / Philosophy of Science): the trichotomy is a philosophical proposition, and the physics content (the adjudication matrix, the scripts) is its testing support.

## Falsification Clause

---

The theorem includes an explicit falsification clause: **if there exists a testable sameness claim whose structural basis cannot be decomposed into the map/projection/invariant three, the theorem is falsified and the classification revised**. The historical candidate for a "fourth"—duality from common boundary conditions (bulk-boundary correspondence)—falls within the projection type in this framework (AdS/CFT is a holographic projection); should future physics provide a genuine fourth, the classification extends accordingly. The value of a completeness proof lies precisely in making such revision precisely describable.

## Conclusion

---

The three unifications are not one viewpoint among many but the three possible structures of sameness claims. The completeness makes the differing evidence regimes of the three classes a structural necessity, and the adjudication matrix the operationalization of that structure. This is a first version; formalization details of the proof (the full statement of functorial decomposition) are left to future work; the falsification clause secures its scientific standing.

## References

1. Halvorson, H. **Scientific Theories** (Oxford, 2019).
2. Butterfield, J. & De Haro, S. et al. On Dualities and Equivalences Between Physical Theories. arXiv:1806.01505 (2018).
3. Weatherall, J. O. Theoretical Equivalence in Physics. arXiv:1810.08192 (2018).
4. De Haro, S. Conceptual Aspects of Gauge/Gravity Duality. arXiv:1509.09231 (2015).
5. De Haro, S. & Butterfield, J. **The Philosophy and Physics of Duality** (Oxford University Press, 2025), 624 pp, open access. arXiv:2508.01616.
6. Book review: Metascience 35, 9 (2026), DOI 10.1007/s11016-026-01101-7.
7. Bern, Z., Carrasco, J. J. & Johansson, H. (double-copy program, review series).
8. Kaluza, T. (1921); Klein, O. (1926) (fibration prototypes).
9. MuningAn. Three Unifications: A Falsifiable Map of Quantum-Gravity Frontiers (2026).

**Statement** The categorical language of this paper serves only as a classification skeleton; all physical content comes from the companion papers and the literature.